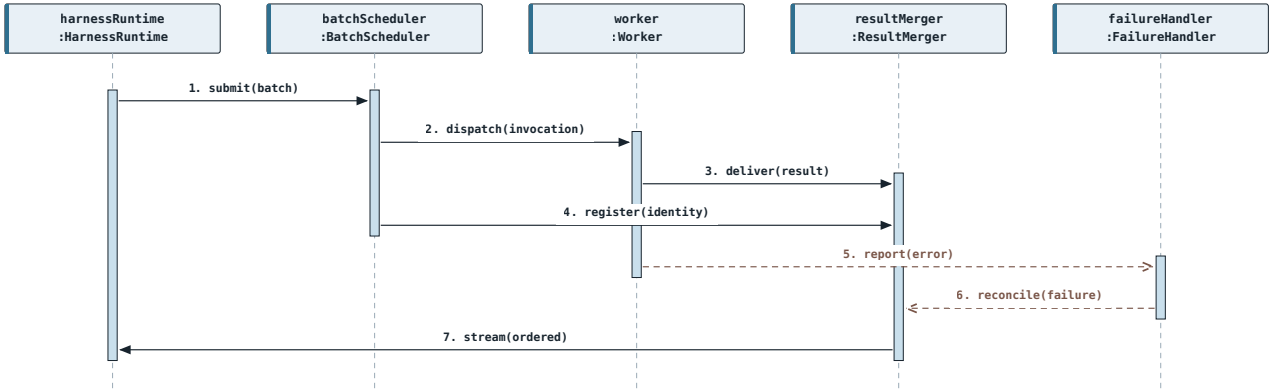
P02 Streaming concurrent batch loop

Primitive / OBJECT INTERACTION / ANALYTICAL COLLABORATION

Overlap output and independent work.

OBJECT INTERACTION

solid arrow: call/message · dashed open arrow: return/failure · analytical order, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	Multiple model, tool, or batch tasks are authorized to run concurrently.
TRIGGER	A batch is dispatched or streamed.
STRUCTURE	Invocation identity, cancellation control, and result correlation.
EFFECT	Streamed or completed results are merged without losing or duplicating ownership.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	The runtime exposes concurrency and a way to await or cancel outstanding work.
ENSURES	Every accepted invocation is linked to its result or explicit failure.
INVARIANT	The harness does not declare completion while unreconciled work remains.
VIOLATION	Partial failure and cancellation produce explicit terminal states and cleanup.

Sources: protocol/pattern_diagram_models.yaml (participants + messages); protocol/harness_pattern_specification.yaml (pointcut + contract).

Analytical collaboration only; no repository execution or message order is asserted.